Quiz Questions

Q1: Database Management Systems (1)
While drawing the ER diagram for a relation “student registers a course”, “registers” should be mentioned _____ 2 rectangles.
A. inside a diamond between
B. over the line connecting
C. inside an oval between
D. inside a rectangle between
Correct Answer: A. inside a diamond between

Q2: Database Management Systems (10)
What is the number of candidate keys for a relation R(A,B,C,D) having functional dependencies {AB->CD, D->B, C->A}?
A. 2
B. 3
C. 4
D. 5
Correct Answer: C. 4

Q3: Database Management Systems (11)
In a relation R(A,B,C,D) having functional dependencies {AB->CD, D->B}, the extraneous attribute in AB->CD is
A. A
B. B
C. C
D. None
Correct Answer: D. None

Q4: Database Management Systems (12)
The highest normal form of a relation R(A,B,C,D) with functional dependencies {A->BCD, BC->AD, D->B} is
A. 1NF
B. 2NF
C. 3NF
D. BCNF
Correct Answer: C. 3NF

Q5: Database Management Systems (13)
The non-prime attributes of the relation R(A,B,C,D,E) with functional dependencies {AE->BC, AC->D, CD->BE, D->E} is/are:
A. A,C,D,E

B. B
C. A,C
D. B,D,E
Correct Answer: B. B

Q6: Database Management Systems (14)
In the relation R(A,B,C,D,E,F) with functional dependencies {AB->C, C->DE, E->F, D->A, C->B}, AB+ is
A. {C}
B. {C,D,E}
C. {C,D,E,F}
D. {A,B,C,D,E,F}
Correct Answer: D. {A,B,C,D,E,F}

Q7: Database Management Systems (15)
What is the relational algebra expression corresponding to the SQL query "SELECT city FROM store"?
A. sigmacity(store)
B. picity(store)
C. sigmastore(city)
D. pistore(city)
Correct Answer: B. picity(store)

Q8: Database Management Systems (16)
Which of the following expressions is used to fetch the details of all stores located in India?
A. sigmacountry='India'(store)
B. picountry='India'(store)
C. sigmastore(country='India')
D. pistore(country='India')
Correct Answer: A. sigmacountry='India'(store)

Q9: Database Management Systems (17)
Which of the following is not a valid schedule of 2 transactions, T1: {R(X), W(X)} and T2: {R(Y), W(Y)}?
A. R(X), W(X), R(Y), W(Y)
B. R(Y), W(Y), R(X), W(X)
C. W(X), W(Y), R(X), R(Y)
D. R(X), R(Y), W(X), W(Y)
Correct Answer: C. W(X), W(Y), R(X), R(Y)

Q10: Database Management Systems (18)
Which of the following pairs of operations conflict in the transactions, T1: {R1(X), W1(X)} and T2: {R2(X), W2(Y)}
A. R1(X),W1(X)
B. R1(X),R2(X)
C. W1(X),R2(X)
D. R1(X),W2(Y)
Correct Answer: C. W1(X),R2(X)

Q11: Database Management Systems (19)
Which of the following is a conflict serializable schedule of the transactions, T1: {R1(X), W1(X), R1(Y)} and T2: {R2(Y), R2(X), W2(X)}?
A. R1(X), R2(Y), R2(X), W1(X), W2(X), R1(Y)
B. R2(Y), R2(X), R1(X), W1(X), W2(X), R1(Y)
C. R1(X), W1(X), R2(Y), R2(X), W2(X), R1(Y)
D. R2(Y), R2(X), R1(X), W2(X), W1(X), R1(Y)
Correct Answer: C. R1(X), W1(X), R2(Y), R2(X), W2(X), R1(Y)

Q12: Database Management Systems (2)
A database designed to store global information contains a field “contact number” in a specific relation. If contact number is to be saved in multiple parts like country code and the actual number, what kind of attribute should be used to represent contact number in the ER diagram?
A. Atomic attribute
B. Composite attribute
C. Multi-valued attribute
D. Derived attribute
Correct Answer: B. Composite attribute

Q13: Database Management Systems (20)
Which of the following statement(s) is/are TRUE?
Consider the following two statements about database transaction schedules:
I. Strict two-phase locking protocol produces conflict serializable schedules which are not recoverable.
II. Timestamp-ordering concurrency control protocol with Thomas’ Write Rule can produce view serializable schedules that are not conflict serializable.
A. I only
B. II only
C. Both I and II
D. Neither I nor II
Correct Answer: B. II only

Q14: Database Management Systems (21)
Which of the following is not a goal of concurrent execution of a transaction?
A. Improved Resource Utilization
B. Improved Throughput
C. Reduced probability of deadlock
D. Reduced waiting time
Correct Answer: C. Reduced probability of deadlock

Q15: Database Management Systems (22)
Which one of the following is true about the concurrency?
A. It ensures that no interference between simultaneous operations on the database
B. It ensures that a transaction allows either all of its operations are executed or none
C. It make sure that database transactions allows any integrity violation in the database.
D. None of the options
Correct Answer: A. It ensures that no interference between simultaneous operations on the database

Q16: Database Management Systems (23)
Which of the following will not allow both dirty reads and non-repeatable reads, but allows phantom reads?
A. Read committed
B. Read uncommitted
C. Repeatable read
D. Serializable
Correct Answer: C. Repeatable read

Q17: Database Management Systems (24)
Which of the following is false?
A. Ensuring durability is the responsibility of recovery management component
B. Ensuring isolation is the responsibility of concurrency management
C. Ensuring atomicity or consistency is the responsibility of recovery management component
D. None of the options
Correct Answer: D. None of the options

Q18: Database Management Systems (25)
Which of the following is true?
A. All the schedules which are allowed under Thomas write rule are also allowed under basic timestamp protocol
B. All the schedules which are allowed under basic timestamp rule are also allowed under multi-version timestamp protocol

C. All the schedules which are allowed under multi-version timestamp protocol are also allowed under Thomas write rule
D. All are true
Correct Answer: B. All the schedules which are allowed under basic timestamp rule are also allowed under multi-version timestamp protocol

Q19: Database Management Systems (26)
Which of the following statement is not correct for serializable of transactions?
A. In a serial schedule, each transaction is independent of others.
B. Every schedule is serializable
C. In non-serial schedule may not always result in an incorrect outcome
D. In non-serial schedule, may result with overlapping of two transaction's executions
Correct Answer: B. Every schedule is serializable

Q20: Database Management Systems (27)
Which of the following is not a recovery technique?
A. Deferred update
B. Immediate update
C. Two-phase commit
D. Recovery management
Correct Answer: C. Two-phase commit

Q21: Database Management Systems (28)
Rollback of transactions is normally used to :
A. Recover from transaction failure
B. Update the transaction
C. Retrieve old records
D. Repeat a transaction
Correct Answer: A. Recover from transaction failure

Q22: Database Management Systems (29)
Which of the following are true about indexes?
A. Indexes will improve the performance of the database table, even it is updated frequently
B. Indexes makes sql server engine's task even tougher, especially with large keys
C. Both
D. None of the options
Correct Answer: B. Indexes makes sql server engine's task even tougher, especially with large keys

Q23: Database Management Systems (3)

What kind of a relation is “a student registers for more than one course for a semester”?
A. one-to-many
B. many-to-many
C. one-to-one
D. None
Correct Answer: B. many-to-many

Q24: Database Management Systems (30)
Consider a B+ tree, the size of the block and node is same as 4 kilobytes. Suppose each index entry is 40 bytes long, find out the height of the tree of a file with one million search key values?
A. 5
B. 4
C. 6
D. 3
Correct Answer: B. 4

Q25: Database Management Systems (31)
The schedule S: r1(X), w3(X), w3(Y), w2(Y), r1(Z), w3(Z), w2(Z) is
A. Both view serializable and conflict serializable schedule
B. View serializable but not conflict serializable schedule
C. Only View serializable
D. Only conflict serializable
Correct Answer: A. Both view serializable and conflict serializable schedule

Q26: Database Management Systems (32)
Which of the following statement(s) is/are not TRUE about the B+ tree, which is used for creating an index of a database table?
A. B+ Tree is a height-balanced tree
B. Non-leaf nodes have pointers to data records
C. Key values in each node are kept in sorted order
D. Each leaf node has a pointer to the next leaf node
Correct Answer: B. Non-leaf nodes have pointers to data records

Q27: Database Management Systems (33)
B+ Trees are normally considered as BALANCED, since
A. Lengths of the paths from the root to all leaf nodes are all equal.
B. Lengths of the paths from the root to all leaf nodes differ from each other by at most
C. Number of children of any two non-leaf sibling nodes differ by at most 1.
D. Number of records in any two leaf nodes differ by at most 1
Correct Answer: A. Lengths of the paths from the root to all leaf nodes are all equal.

Q28: Database Management Systems (34)
Primary indexes of multilevel indexes are created usually based on its first level and it is normally referred as
A. zero level of multilevel index
B. third level of multilevel index
C. second level of multilevel index
D. first level of multilevel index
Correct Answer: C. second level of multilevel index

Q29: Database Management Systems (35)
Which of the following is/are correct?
A. B-trees are for storing data on disk and B+ trees are for main memory.
B. Range queries are faster on B+ trees.
C. B-trees are for primary indexes and B+ trees are for secondary indexes.
D. The height of a B+ tree is independent of the number of records.
Correct Answer: B. Range queries are faster on B+ trees.

Q30: Database Management Systems (36)
Find the maximum and minimum number of keys in the internal nodes of B – Tree if the order of the node is four,
A. 4,2
B. 3,1
C. 4,0
D. 4,3
Correct Answer: B. 3,1

Q31: Database Management Systems (37)
Which of the following statement(s) are incorrect
A. In non-relational databases, schemas are need to be defined before adding data
B. NoSQL databases allows the insertion of data without a predefined schema
C. NewSQL databases allows the insertion of data without a predefined schema
D. All the statements
Correct Answer: A. In non-relational databases, schemas are need to be defined before adding data

Q32: Database Management Systems (38)
NoSQL databases is used mainly for handling large volumes of _____ data.
A. Unstructured
B. Structured
C. Semi-structured

D. All the options
Correct Answer: A. Unstructured

Q33: Database Management Systems (39)
Which of the following are the NoSQL databases?
A. Key-value
B. Wide-column
C. Document
D. All the options
Correct Answer: A. Key-value

Q34: Database Management Systems (4)
If an athlete can participate in a maximum of 3 activities, what is cardinality ratio of ATHLETE:ACTIVITY in PARITICIPATE?
A. 1:n
B. 2:1
C. 1:3
D. n:1
Correct Answer: C. 1:3

Q35: Database Management Systems (40)
Consider the following schedule and state that which of the following statement(s) is (are) true?
S : r1(x),w1(x),r2(x),r1(y),w1(y),w2(x),r2(y), c1, c2;
I: S is recoverable.
II: If the order of w1(x) and w2(x) are interchanged, then S will be a non-recoverable schedule.
A. Only I
B. Only II
C. Both I and II
D. None of the options
Correct Answer: A. Only I

Q36: Database Management Systems (41)
For the given schedule, which of the following statements is true about the schedule S?
S : r1(y), r2(x), r2(y), r3(y), w2(x), w1(y), w3(x), r1(x)
A. S is the conflict serializable but not view serializable
B. S is neither conflict serializable nor view serializable
C. S is both conflict serializable as well as view serializable
D. S is view serializable but not conflict serializable
Correct Answer: C. S is both conflict serializable as well as view serializable

Q37: Database Management Systems (42)
Which of the following is not an entity set in a SHOPPING mini world?
A. Customers
B. Stock
C. Products
D. ProductName
Correct Answer: D. ProductName

Q38: Database Management Systems (43)
Identify the type of database system required to store a research document.
A. Multimedia Database
B. Geographical Information System
C. Data Warehouse
D. Textual Database
Correct Answer: D. Textual Database

Q39: Database Management Systems (44)
In a Bank scenario, the window visible to a customer while doing transactions via net banking is referred to as -----
A. The database state
B. The Schema
C. The View
D. Data dictionary
Correct Answer: C. The View

Q40: Database Management Systems (45)
A bank operation accommodates several operations in a day. A snapshot taken at 4 p.m to know the transactions currently happening is referred to as----------
A. Database Schema
B. Database state
C. Abstraction
D. Catalogue
Correct Answer: B. Database state

Q41: Database Management Systems (46)
Which of the following is a possible schema construct in a supermarket mini world?
A. Products
B. inventory
C. sales
D. All the options

Correct Answer: D. All the options

Q42: Database Management Systems (47)
Suppose you are assigned to do a standalone database application, what type of DBMS architecture is required?
A. Single tier architecture
B. Two tier
C. Three tier
D. None of the mentioned
Correct Answer: A. Single tier architecture

Q43: Database Management Systems (48)
Identify the possible entity sets of a Cafeteria Data base.
A. Customers, menu manager, meal Delivery
B. Customers, menuManager, itemsProcurement, mealDelivery,billGeneration
C. Customers, menu manager, Bill generation
D. FoodMenu
Correct Answer: B. Customers, menuManager, itemsProcurement, mealDelivery,billGeneration

Q44: Database Management Systems (49)
The attribute Address could be structured as door no, street_name, City and Zipcode. This type of attribute is called ---------
A. Simple attribute
B. Multivalued attribute
C. Composite attribute
D. Derived attribute
Correct Answer: C. Composite attribute

Q45: Database Management Systems (5)
For a relation R(A,B,C,D,E,F,G), the candidate keys are AB, AC, and DE. Which of the following is not a proper subset of the candidate key?
A. BC
B. DE
C. C
D. AD
Correct Answer: B. DE

Q46: Database Management Systems (50)
The function that an entity Instructor plays in a relationship teaches refers to the -------
A. Participation

B. Role
C. Instance
D. Position
Correct Answer: B. Role

Q47: Database Management Systems (51)
Which of the following can be a multivalued attribute?
A. Phone_number
B. Name
C. Date_of_birth
D. All the options
Correct Answer: A. Phone_number

Q48: Database Management Systems (52)
Identify the key attribute of the entity PROJECT( ProjectId, Projectname, ProjLocation)
A. ProjectId, ProjectName
B. ProjectName
C. ProjLocation
D. ProjectId
Correct Answer: D. ProjectId

Q49: Database Management Systems (53)
The attribute experience is calculated from the attribute dateOfJoining. What is the type of the attribute?
A. Derived attribute
B. Descriptive attribute
C. Simple attribute
D. Multivalued attribute
Correct Answer: A. Derived attribute

Q50: Database Management Systems (54)
SLOT is a weak entity of the University schema. Which of the statements is not true?
A. It requires a identifying relationship TAKES
B. It has a identifying owner entity TIMETABLE
C. It should have total participation with respect to the entity TAKES
D. There is no existence dependency of the entity SLOT with respect to the entity TIMETABLE
Correct Answer: D. There is no existence dependency of the entity SLOT with respect to the entity TIMETABLE

Q51: Database Management Systems (55)

ORDERS is an entity set with attributes (orderId, items, price) and SHIPPING entity is a specialized version of the ORDERS entity. Which of the following are the attribute set of SHIPPING entity?
A. (OrderId, items, price)
B. (Orderid, Shippingid)
C. (OrderId, items, price, ShippingId,FromLocation,ToLocation)
D. (OrderId, Items, ShippingId,FromLocation,ToLocation)
Correct Answer: C. (OrderId, items, price, ShippingId,FromLocation,ToLocation)

Q52: Database Management Systems (56)
If an attribute is added to the superclass entity, the condition for propagating this change in subclass entity is
A. Attribute defined membership
B. Predicate defined membership
C. Both Attribute defined and Predicate defined membership
D. Overlapping participation
Correct Answer: C. Both Attribute defined and Predicate defined membership

Q53: Database Management Systems (57)
“Customer buys Products”. Identify the relationship set.
A. Customer
B. Products
C. Buys
D. Customer and Products
Correct Answer: C. Buys

Q54: Database Management Systems (58)
The entity set MovieZone is classified as Hollywood and Bollywood. This process is called ------
A. Generalization
B. Specialization
C. Inheritance
D. Constraint generalization
Correct Answer: B. Specialization

Q55: Database Management Systems (59)
Identify the simple attribute.
A. name
B. grade
C. Color
D. VehicleRegistrationNumber
Correct Answer: A. name

Q56: Database Management Systems (6)
If a relational schema "R" is in BCNF, which of the following statements is true?
A. R is in 4NF
B. R is in 2NF but not in 3NF
C. R is not in 1NF
D. R is in 2NF and 3NF
Correct Answer: D. R is in 2NF and 3NF

Q57: Database Management Systems (60)
An entity set DOCTOR has a total participation on the relationship DIANOSES that connects it to the entity set PATIENTS. What would be the minimum cardinality constraint on the relationship?
A. 1:1
B. M:N
C. 1:N
D. N:1
Correct Answer: B. M:N

Q58: Database Management Systems (61)
An entity set SHIPPING has the attributes (shippingId, ShipLocations). Which of the following is true?
A. ShippingId is the primary key
B. ShipLocations is the primary key
C. ShippingId, ShipLocations is a composite key
D. None of the options
Correct Answer: C. ShippingId, ShipLocations is a composite key

Q59: Database Management Systems (62)
SENSE, SELECT, SCOPE are the entity sets with common characteristics. Which process can define an entity set called DEPARTMENTS?
A. Specialization
B. Generalization
C. Top down approach
D. Inheritance
Correct Answer: B. Generalization

Q60: Database Management Systems (63)
In a share market application, an entity set INVESTOR can buy or sell SHARES. This refers to --------
A. Relationship

B. Role definition
C. Recursive relationship
D. Participation
Correct Answer: C. Recursive relationship

Q61: Database Management Systems (64)
Which one of the following events in a database system can lead to an irrecoverable mistake?
A. A transaction reads a data item after it is written by an uncommitted transaction
B. A transaction reads a data item after it is written by a committed transaction
C. A transaction reads a data item after it is read by an uncommitted transaction
D. A transaction writes a data item after it is read by an uncommitted transaction
Correct Answer: A. A transaction reads a data item after it is written by an uncommitted transaction

Q62: Database Management Systems (65)
Consider the table R with attributes A, B and C. The functional dependencies that hold on R are : A → B, C → AB. Which of the following statements is/are True ? I. The decomposition of R into R1(C, A) and R2(A, B) is lossless. II. The decomposition of R into R1(A, B) and R2(B, C) is lossy.
A. Only I
B. Only II
C. Both I and II
D. Neither I nor II
Correct Answer: C. Both I and II

Q63: Database Management Systems (66)
Which of the following is correct with respect to Two phase commit protocol?
A. Ensures serializability
B. Prevents Deadlock
C. Detects Deadlock
D. Recover from Deadlock
Correct Answer: A. Ensures serializability

Q64: Database Management Systems (67)
Consider a B+-tree in which the maximum number of keys in a node is 7. What is the minimum number of keys in any non-root node?
A. 3
B. 1
C. 2
D. 4
Correct Answer: A. 3

Q65: Database Management Systems (68)
A file is structured in such a way that the ordering of data records in certain indexes is the same as or similar to the ordering of data entries. That index is then called
A. Clustered
B. Sparse
C. Dense
D. Unclustered
Correct Answer: A. Clustered

Q66: Database Management Systems (69)
The physical position of a record is calculated by a formula that translates a file key into a position for the record is
A. B-Tree file
B. Hashed file
C. Indexed file
D. Sequential file
Correct Answer: B. Hashed file

Q67: Database Management Systems (7)
SELECT _____ destination FROM flight;
Fill the blank in the above SQL query with an appropriate keyword to display all non-repeating values of the column “destination” only once and values that occur once as such.
A. DIFFERENT
B. DISTINCT
C. ALL
D. UNIQUE
Correct Answer: B. DISTINCT

Q68: Database Management Systems (70)
Relation R has six attributes ABCDEF. Fields of R contain only atomic values. F = { A -> BC, B -> CFH, E -> A, F -> E} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. How many candidate keys does the relation R have?
A. 5
B. 6
C. 3
D. 4
Correct Answer: D. 4

Q69: Database Management Systems (71)
Consider the given functional dependencys (FDs) of a relation R has six attributes ABCDEF. Fields of R contain only atomic values. F = { A -> BC, B -> CFH, E -> A, F -> E} is a set of functional dependencies (FDs) so that F+ is exactly the set of FDs that hold for R. The relation R is in
A. 2NF, but not in 3NF.
B. 3NF, but not in BCNF.
C. 1NF, but not in 2NF.
D. BCNF.
Correct Answer: C. 1NF, but not in 2NF.

Q70: Database Management Systems (72)
Consider the relation scheme R = {A, B, C, D, E, F, G} and the set of functional dependencies {{A, B} -> {C}, {B} -> {E, F}, {A, D} ->{G}} on R. What is the key for R?
A. {A, B, C, D, G}
B. {A, D}
C. {A, B}
D. {A, B, D}
Correct Answer: D. {A, B, D}

Q71: Database Management Systems (73)
The maximum number of superkeys for the relation schema R(A, B, C, D, E, F, G, H) with H as the key is
A. 64
B. 128
C. 7
D. 256
Correct Answer: B. 128

Q72: Database Management Systems (74)
Select the 'False' statement from the following statements about Normal Forms:
A. Lossless preserving decomposition into 3NF is always possible
B. Any Relation with two attributes is in BCNF
C. Lossless preserving decomposition into BCNF is always possible
D. BCNF is stronger than 3NF
Correct Answer: C. Lossless preserving decomposition into BCNF is always possible

Q73: Database Management Systems (75)
A B-tree of order 4 is constructed by 8 successive insertions from scratch. What is the maximum number of operations for node splitting that can take place?
A. 3

B. 4
C. 5
D. 6
Correct Answer: A. 3

Q74: Database Management Systems (76)
The 'Durability' property among the ACID properties of a transaction requires that the modifications made by a successful transaction to the database survive.
A. Except in case of an Operating System crash
B. Always, even if there is a failure of any kind
C. Except in case of a power failure
D. Except in case of a Disk crash
Correct Answer: B. Always, even if there is a failure of any kind

Q75: Database Management Systems (77)
Consider the following schema:
Emp (Empcode, Name, Sex, Salary, Deptt)
A simple SQL query is executed as follows:
SELECT Deptt FROM Emp
WHERE sex = 'M'
GROUP by Dept
Having avg (Salary) > {select avg (Salary) from Emp}
The output will be?
A. Average salary of male employee is more than the average salary of the organization
B. Average salary of male employee is equal to the average salary of the organization
C. Average salary of male employee is the average salary of the organization
D. Average salary of male employee is less than the average salary of the organization
Correct Answer: A. Average salary of male employee is more than the average salary of the organization

Q76: Database Management Systems (78)
Consider the following query :
SELECT E.eno, COUNT(*)
FROM Employees E
GROUP BY E.eno
If an index on eno is available, the query can be answered by scanning only the index if
A. index can be hash or B+ tree and clustered or non-clustered
B. the index is only hash and clustered
C. the index is only B+tree and clustered
D. index can be hash or B+ tree and clustered
Correct Answer: A. index can be hash or B+ tree and clustered or non-clustered

Q77: Database Management Systems (79)
Which level of locking provides the highest degree of concurrency in a relational data base?
A. Table
B. Row
C. Page
D. Page, table and row level locking allow the same degree of concurrency
Correct Answer: B. Row

Q78: Database Management Systems (8)
SELECT * FROM player WHERE num_of_matches>100 AND country='India';
What fields will be outputted by the above SQL query?
A. player
B. num_of_matches
C. num_of_matches, country
D. All fields of player relation
Correct Answer: D. All fields of player relation

Q79: Database Management Systems (9)
Which of the following SQL queries can be used to change "Mishra" to "Singh" in the "LastName" column in the People relation?
A. UPDATE People SET LastName='Mishra' INTO LastName='Singh'
B. UPDATE People SET LastName='Singh' WHERE LastName='Mishra'
C. MODIFY People SET LastName='Mishra' INTO LastName='Singh'
D. MODIFY People SET LastName='Singh' WHERE LastName='Mishra'
Correct Answer: B. UPDATE People SET LastName='Singh' WHERE LastName='Mishra'